

Pritam Dash

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EXPERIENCE

Voice Model Developer, Contract

Feb 2026 – May 2026

Trelis Research

Remote

- Fine-tuned Whisper-large-v3 for Greek and Greek-English speech to **~2.85% CER**, beating ElevenLabs Scribe v2 (~5.85%)
- Cut real-world meeting CER from ~19.9% to **~11.9%** (~40% relative improvement) via data mix and recipe tuning
- Trained Greek TTS – Piper (VITS) at **28× real-time on CPU** and Orpheus-3B at **83.5× on H100**, both <5% CER
- Deployed Whisper-large-v3 on vLLM at **~666× real-time throughput** on a single H100 for production serving
- Contributed to an internal speech ML platform spanning training, evaluation, and serving of ASR and TTS models

ML Engineering Advisor

Oct 2025 – Feb 2026

HeyMaya (Meshly Technologies)

Remote

- Built WhatsApp conversational AI for HR automation, serving **150K+ users** with **200ms** response times
- Trained and deployed LLM on AWS SageMaker reaching **97% accuracy** across **10K+ daily conversations**
- Reduced monthly cloud infrastructure costs by **40%** through model optimization

SDE Intern

Jan 2025 – Jun 2025

Hero MotoCorp

Gurugram, India

- Automated invoice reconciliation for 500+ monthly invoices, reducing manual review time by **60%**
- Optimized SQL queries and database indexing, improving dashboard load times from **8.2s to 1.3s**

Data Analyst Intern

May 2024 – Jul 2024

Ennovations Techserve

Noida, India

- Automated ETL workflows for 50K+ records using Zoho Analytics, eliminating **80%** of manual data entry

Technical Head

Dec 2023 – Jun 2025

AIEC, IITM

New Delhi, India

- Directed 46-hour hackathon for **1,200+ participants**; ran 8 ML workshops for **300+ students**

TECHNICAL SKILLS

Languages: Python

ML / Deep Learning: PyTorch, Hugging Face Transformers, PEFT, TRL, Unsloth

Speech AI: ASR, TTS, streaming voice agents, audio post-processing, neural audio codecs

LLM Fine-Tuning: LoRA / QLoRA, SFT, GRPO

Inference & Serving: vLLM, llama.cpp, CTranslate2, INT4 / INT8 quantization

Evaluation Metrics: WER, CER, CMER, UTMOS, RTF, TTFA

Cloud & Deployment: AWS (SageMaker, EC2), Docker, FastAPI, WebSocket

PROJECTS

Real-Time Conversational Voice AI | Streaming voice agent with RL-tuned neural TTS (Orpheus 3B)

- Built ASR + LLM + TTS streaming pipeline with **sub-800ms time-to-first-audio** on self-hosted GPUs
- Fine-tuned 3B-parameter neural TTS via reinforcement learning (UTMOS reward) for higher naturalness

Self-Hosted Voice Agent | Single-GPU real-time conversational AI (CSM-1B)

- Shipped real-time voice agent on a single GPU with **1.3s latency**, barge-in, and natural-prosody audio
- Fine-tuned 1B-parameter conversational TTS while preserving base-model prosodic priors

Multilingual ASR Fine-Tuning (Hindi) | Parameter-efficient speech recognition (Whisper large-v3-turbo)

- Reduced Hindi WER from **35.6% to 22.3%** (FLEURS) by fine-tuning 3% of an 800M-parameter ASR model
- Built ASR pipeline: data prep, LoRA training, INT8 quantization, faster-whisper deployment

HeyMaya | Production LLM platform for HR automation

- Trained conversational AI on 50K+ HR dialogues; deployed to production serving 150K+ active users
- Optimized inference pipeline achieving 200ms p95 latency for real-time conversations on mobile networks

Indian Classical Music Separation | Deep learning for audio source separation

- Trained audio source separation model achieving 12.45 dB SDR, outperforming Meta's Demucs by 53%
- Designed custom loss function for harmonically complex Indian instruments, trained on consumer-grade GPU

Research Assistant API | Semantic search over academic papers

- Built semantic search API over research papers improving LLM answer accuracy by 60% through context retrieval
- Implemented vector database with FAISS enabling sub-100ms similarity search across 10K+ documents

ACHIEVEMENTS & LEADERSHIP

1st Place AI/ML Hackathon, 2024 | **Speaker** ED Talk World Conference, 2024 | **Technical Head** AIEC, IITM

EDUCATION

Maharshi Dayanand University - Bachelor of Computer Applications, 2025

CBSE - Senior Secondary, 84%, 2021

CBSE - Secondary, 92%, 2019